

# LQG Notes

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# 1 Refined Algebraic Quantization

## *A Quantization Programme for Theories With Constraints*

As we are all well aware, the whole quantization programme of QFT is based on the so-called Wightman Axioms. We can loosely summarize this procedure into four steps: fixing a background metric  $\eta_{\mu\nu}$ , finding its corresponding symmetric group  $\mathcal{P}$ , formalizing the commutation relations and finally defining the vacuum state. From this, one can construct the full Fock space, which is the space of physical states. However, if one is to generalize this procedure to General Relativity, one will immediately run into trouble for failing to fix a background. For this and many other reasons, we must search for another way to quantize General Relativity.

Luckily, GR can be formulated as a constrained Hamiltonian system with first class constraints. The quantization of such a system had been explored by Dirac and many others and is known by the name of **Refined Algebraic Quantization**. This note is written in the aim of refining my understanding on this topic.

For any system, we start with a phase space  $\mathcal{M}$ . It has a set of constraints  $C_I$  (In the case of GR, these constraints reflects the degrees of freedom in foliation and choosing spatial coordinates on hypersurfaces.) and a Hamiltonian  $\mathcal{H}$ .

Here let's stop to consider what a phase space with constraints will look like.  $C_I$  limits  $\mathcal{M}$  to a hypersurface  $\bar{\mathcal{M}}$ . It is also the generator of gauge transformations that leaves  $\bar{\mathcal{M}}$  invariant. Therefore if we consider a point  $m \in \bar{\mathcal{M}}$ , a gauge  $C$  generates an orbit  $[m]$  on  $\bar{\mathcal{M}}$ . We can now assert that the physical phase space is the equivalence class of  $[m]$ .

For simplicity, we choose a polarization on  $\mathcal{M}$  such that  $\mathcal{M}$  is a cotangent bundle, and the configuration space  $\mathcal{C}$  as its base manifold:  $\mathcal{M} = T^*\mathcal{C}$ . We will also assume there exist a lagrangian  $\mathcal{L}$  and vanishing poisson brackets on  $\mathcal{C}$ . With these assumptions, we can always consider the fiber of  $\mathcal{M}$ , that is the momentum space. Then we compute the poisson bracket between the momentum and a function  $f(q) \in C^\infty(\mathcal{C})$ :  $\{p, f\}(q)$ . These brackets have the same structure as a derivative of  $f$ :  $\{p, \cdot\}(f)(q)$ . It is easy to recognize this object as a vector on configuration space  $v \in V^\infty(\mathcal{C})$ , therefore we denote  $v_p[f] := \{p, f\}(q)$ , a vector field generated by  $p$ . This vector field no doubt carries the physical dynamics of the system and preserves the symplectic structure. We can also consider the Lie algebra  $C^\infty(\mathcal{C}) \times V^\infty(\mathcal{C})$  defined by  $[(f, v), (f', v')] = (v[f'] - v'[f], [v, v'])$ , choosing the preferred vector field  $v_p$  gives us a subalgebra  $(f, v_p)$  denoted by  $\mathcal{B}$ .

Equipped with this algebra  $\mathcal{B}$ , we can construct the corresponding Hilbert space. This step is similar to the process of constructing Fock space with Poincare group in QFT, but here we won't result in a space of physical states, rather a kinematical space. We need a irreducible *-representation*  $\pi : \mathcal{B} \rightarrow L(\mathcal{H}_{kin})$ . That is to say,  $\pi$  is a linear operator on  $\mathcal{H}_{kin}$  such that  $\pi(a)\pi(b) = \pi(b)\pi(a)$  and  $\forall a, b \in \mathcal{B}$ ,

$$\begin{cases} \pi(a)^\dagger = \pi(a^*), \\ [\pi(a), \pi(b)] = i\hbar\pi([a, b]). \end{cases}$$

As stated by the **Groenewald-van Hove Theorem**, it is impossible to faithfully represent the full poisson algebra.

One can usually describe a Hilbert space in the form  $L_2(\bar{\mathcal{C}}, d\mu)$ , where  $\bar{\mathcal{C}}$  is an extension of the configuration space, and  $d\mu$  a measure.

Even though Groenewald-van Hove Theorem limits the numbers of representations we can take, there is still plenty of reps for us to choose from. We want our reps to also represent the constraints and Hamiltonians into operators. But these corresponding operators are often unbounded and therefore not well-defined on all of  $\mathcal{H}_{kin}$ . We need a dense subspace which is close to Hamiltonians and constraints and can avoid infinities. We accomplish this by choosing a space of rapid decrease smooth functions  $\mathcal{D}_{kin}$ . By this, we mean  $\mathcal{D}_{kin}$  has a finer topology than  $\mathcal{H}_{kin}$ , thus a topological inclusion

$$\mathcal{D}_{kin} \hookrightarrow \mathcal{H}_{kin}.$$

We now impose our constraints. The quantum version of the constraint equation  $\hat{C}_I l = 0$  is basically requiring  $l$  to be invariant under gauge transformations. Because of this, when we compute the total probability of a physical state  $l$ , we should integrate  $l$  along its gauge orbit:

$$\|l\|^2 = \int_{-\infty}^{+\infty} |l|^2 d\mu$$

which obviously diverges. This means physical states, namely states that solve the constraint equation cannot lie in  $\mathcal{H}_{kin}$ . Observe that  $\hat{C}_I$  has continuous spectra; for self-adjoint operators like this, **Gel'fand-Maurin Theorem** guides us to find its eigenvectors in the algebraic dual space of a nuclear space of the Hilbert space, which is exactly  $\mathcal{D}_{kin}^*$ . This by definition is a space of linear functionals, and has pointwise divergence. Again  $\mathcal{H}_{kin}$  has a finer topology than  $\mathcal{D}_{kin}^*$ . We now have the **Gel'fand triple**:

$$\mathcal{D}_{kin} \hookrightarrow \mathcal{H}_{kin} \hookrightarrow \mathcal{D}_{kin}^*.$$

We now define the dual of the constraint operator as

$$[\hat{C}'_I l](f) := l(\hat{C}_I^\dagger f)$$

where  $l \in \mathcal{D}_{kin}^*$  and  $f \in \mathcal{D}_{kin}$ . It is a natural extension of the following relation in  $\mathcal{H}_{kin}$ :

$$[\hat{C}'_I l](f) = \langle \hat{C}_I l, f \rangle_{kin} = \langle l, \hat{C}_I^\dagger f \rangle_{kin} = l(\hat{C}_I^\dagger f).$$

Now the constraint equation becomes

$$[\hat{C}'_I l](f) = l(\hat{C}'_I f) = 0 \quad \forall f \in \mathcal{D}_{kin}.$$

elements of  $\mathcal{D}_{kin}^*$  that solve this equation form a subspace  $\mathcal{D}_{phys}^* \subset \mathcal{D}_{kin}^*$ .

Quantum theory comes with anomalies. Constraints come with an algebra and a structural function  $\{C_I, C_J\} = f_{IJ}^K C_K$ . After quantizing, the algebra becomes  $[\hat{C}_I, \hat{C}_J] = i\hbar \hat{f}_{IJ}^K \hat{C}_K = i\hbar \{[\hat{C}_K, \hat{f}_{IJ}^K] + \hat{f}_{IJ}^K \hat{C}_K\}$ . Any physical state would now also solve the equation  $[\hat{C}_K, \hat{f}_{IJ}^K]l = 0$ , which itself may not be a constraint. Therefore the quantized theory may have less physical degrees of freedom than the classical theory.

As we've discussed,  $\mathcal{H}_{kin} \cap \mathcal{D}_{phys}^* = \emptyset$ , which is annoying because we can't equip  $\mathcal{D}_{phys}^*$  with a scalar product. To resolve this, we introduce **Dirac Observables**. A *strong* dirac observable is an operator on  $\mathcal{H}_{kin}$  that commutes with all constraints and are densely defined on  $\mathcal{H}_{kin}$ . A *weak* dirac observable only commutes with the constraint on the constraint hypersurface:  $[\hat{O}, \hat{C}_I]_{|C_J=0} = 0$ . These observables also have their natural dual  $\hat{O}'$  on  $\mathcal{D}_{kin}^*$ , then a weak dirac observable is an operator that leaves the physical space invariant:  $\hat{O}'\mathcal{D}_{phys}^* \subset \mathcal{D}_{phys}^*$ .

Now we can finally define a inner product on a subset  $\mathcal{H}_{phys} \subset \mathcal{D}_{phys}^*$  as a positive definite sesquilinear form with respect to which  $\hat{O}'$  becomes self-adjoint operators, that is  $\hat{O}' = (\hat{O})^*$ . We also have  $[\hat{O}'_1, \hat{O}'_2] = ([\hat{O}_1, \hat{O}_2])'$ . The commutator on the RHS is defined on  $\mathcal{H}_{kin}$ , but the LHS is defined on  $\mathcal{H}_{phys}$ . This is a nice relation because the commutation relations on  $\mathcal{H}_{kin}$  are automatically tranferred to  $\mathcal{H}_{phys}$ . Here we still need a dense subspace  $\mathcal{D}_{phys} \subset \mathcal{H}_{phys}$  where we have well defined operators for reasons explained above. We have finally resulted in a second Gel'fand triple:

$$\mathcal{D}_{phys} \hookrightarrow \mathcal{H}_{phys} \hookrightarrow \mathcal{D}_{phys}^*.$$

This is where the physics happen.

The logic here is that when we take our dirac observables and force them to be self-adjoint, we hope these restrictions are powerful enough to limit the form of the inner product.

In very fortunate cases where constraints are mutually self-adjointing operators, namely  $[\hat{C}_I, \hat{C}_J] = 0$ , all constraints share the same eigenstate, therefore by spectrum theorem we can decompose  $\mathcal{H}_{kin}$  into

$$\mathcal{H}_{kin} = \int_S^\oplus d\mu(\lambda) \mathcal{H}_\lambda$$

over the spectrum  $S$  of the constraint algebra. Then it's obvious that  $\mathcal{H}_{phys} = \mathcal{H}_0$ .

The whole process is basicly  $\mathcal{H}_{kin} \rightarrow \mathcal{D}_{kin} \rightarrow \mathcal{D}_{kin}^* \rightarrow \mathcal{D}_{phys}^* \rightarrow \mathcal{H}_{phys} \rightarrow \mathcal{D}_{phys}$ .

The last step is to make sure this quantum theory actually has the correct classical limit. In other words, we have to make sure that

$$\left| \frac{\langle \psi_{[m]}, \hat{O}' \psi_{[m]} \rangle_{phys}}{O(m)} - 1 \right| \ll 1, \quad \left| \frac{\langle \psi_{[m]}, (\hat{O}')^2 \psi_{[m]} \rangle_{phys}}{(\langle \psi_{[m]}, \hat{O}' \psi_{[m]} \rangle_{phys})^2} - 1 \right| \ll 1.$$

This means the dirac observables have correct expectation values and their relative fluctuations are small. Only then we are sure we have constructed a quantized theory that admits the correct classical limit.

In the next post, we will discuss how these abstract mathematical procedures can be used to really QUANTIZE a theory with constraints.

**Reference:**

- [arXiv:gr-qc/0210094v1](https://arxiv.org/abs/gr-qc/0210094v1)

## 2 Some Trivial Examples of RAQ

*Including momentum constraint, angular momentum constraint, etc.*

In this post we put forward four examples of quantizing systems with constraints using the RAQ programme. In each case our final goal is to derive the dirac observables and the physical hilbert space.

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### 2.1 Momentum Constraint

Consider the phase space  $\mathcal{M} = T^*\mathbb{R}^2$  with poisson brackets among  $q^a$  and  $p_a$ . Our constraint is  $C = p_1$ , and we choose  $\mathcal{H}_{kin} = L_2(\mathbb{R}^2, d^2x)$ ,  $\mathcal{D}_{kin} = \mathcal{S}(\mathbb{R}^2)$  and  $\mathcal{D}_{kin}^* = \mathcal{S}^*(\mathbb{R}^2)$ .

By the constraint  $C = p_1$ , we have the constraint equation  $l(\hat{C}^\dagger \phi) = 0$ , where  $l \in \mathcal{D}_{kin}^*$  and  $\phi \in \mathcal{D}_{kin}$ . Working in the momentum representation, we can write the constraint equation as

$$l(\hat{p}_1 \phi(\hat{p}_1, \hat{p}_2)) = 0$$

where we can clearly see that a function of the form  $l = \delta(p_1)f(p_2)$  solves the equation. We construct the new inner product as  $\langle l_{f_1}, l_{f_2} \rangle := l_1[\phi_2]$ , therefore we have

$$l_1[\phi_2] = \int \int \delta(p_1) f_1^*(p_2) \phi_2(p_1, p_2) dp_1 dp_2 = \int f_1^*(p_2) \phi_2(p_2) dp_2.$$

which is a perfectly well-defined inner product, therefore  $\mathcal{H}_{phys} = L_2(\mathbb{R}, dx_2)$ .

From the constraint it's also trivial to find the dirac observables to be  $q^2, p_2$  because these operators must commute with the constraint, which in this case is  $p_1$ .

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### 2.2 Angular Momentum Constraint

We consider the phase space to be  $\mathcal{M} = T^*\mathbb{R}^3$  with poisson brackets among  $q^a$  and  $p_a$ . Our constraints are  $C_a = \epsilon_{abc} x_b p_c$ , and we choose  $\mathcal{H}_{kin} = L_2(\mathbb{R}^3, d^3x)$ ,  $\mathcal{D}_{kin} = \mathcal{S}(\mathbb{R}^3)$  and  $\mathcal{D}_{kin}^* = \mathcal{S}^*(\mathbb{R}^3)$ .

The constraint algebra is simply the angular momentum algebra, so we can identify  $\hat{C}_a \equiv \hat{L}_a$ . The constraint equation is now  $\hat{L}_a \Psi = 0$ . By expanding it into spherical harmonics, we have

$$\Psi = \sum_{l=0}^{\infty} \sum_{m=-l}^l R_{lm}(r) Y_l^m(\theta, \phi).$$

The constraint equation implies  $\hat{L}^2 \Psi = 0$ , but from QM we know that  $\hat{L}^2 \Psi = l(l+1)\hbar^2 Y_l^m$ , thus  $l = 0$ . Our physical states are then  $\Psi = \phi(r)$ , which can be perfectly normalized in  $\mathcal{H}_{kin}$

because  $SO(3)$  is compact and the integral on the group manifold gives a finite result. Therefore, we have

$$\mathcal{H}_{phys} = L_2(\mathbb{R}, r^2 dr) \subset \mathcal{H}_{kin}.$$

Now we find the dirac observables. We'll have to find operators that are self-adjoint under the physical inner product induced by  $\mathcal{H}_{kin}$ . Notice that if we define  $\hat{p} = -i\hbar r^{-1} \frac{d}{dr} r$ , such a inner product can be written as

$$\begin{aligned} \langle \phi, \hat{p}\psi \rangle &= \int \phi^* \hat{p}\psi r^2 dr = \int \phi^* \left( -i\hbar \frac{1}{r} \frac{d}{dr} (r\psi) \right) r^2 dr \\ &= -i\hbar \int r \phi^* (\partial_r (r\psi)) dr \\ &= (-i\hbar (r\phi^*)(r\psi)) \Big|_{-\infty}^{+\infty} + i\hbar \int \partial_r (r\phi^*) (r\psi) dr \\ &= \int (\hat{p}\phi)^* \psi r^2 dr = \langle \hat{p}\phi, \psi \rangle. \end{aligned}$$

Similarly the operator  $\hat{r} := r$  is also a dirac observable, completing the programme.

### 2.3 Relativistic Particle

From the form of the Lagrangian

$$L = -m \sqrt{-\eta_{\mu\nu} \dot{q}^\mu \dot{q}^\nu}$$

and the definition of conjugate momenta  $p_\mu = \partial L / \partial \dot{q}^\mu$ , we can surprisingly see

$$\frac{\partial L}{\partial \dot{q}^\mu} = m \frac{\dot{q}_\mu}{\sqrt{-\eta_{\mu\nu} \dot{q}^\mu \dot{q}^\nu}} = p_\mu \implies \eta^{\mu\nu} p_\mu p_\nu + m^2 = 0.$$

which is the mass-shell constraint. In this case, the constraint came directly from the lagrangian. We shall later see this is the result of the disappearing Hamiltonian.

Next we prove the action  $S = \int dt L$  to be  $\text{Diff}(\mathbb{R})$  invariant. Consider the reparameterization  $t \mapsto \varphi(t)$ . With this new time variable, the action is now

$$S' = \int -m \sqrt{-\eta_{\mu\nu} \frac{d\dot{q}^\mu}{d\varphi} \frac{d\dot{q}^\nu}{d\varphi}} d\varphi = \int -m \sqrt{-\eta_{\mu\nu} \frac{d\dot{q}^\mu}{dt} \frac{d\dot{q}^\nu}{dt}} \dot{\varphi} \dot{\varphi}^{-1} dt = S.$$

Under reparameterization, although  $\dot{q}^\mu$  is variant,  $p_\mu$  stayed constant. This means the mapping from velocity space to momentum space is not one-to-one, namely every momentum corresponds to a ray in velocity space. Therefore we need to have some kind of constraint on  $\mathcal{M}$ , limiting the physical momentum to a hypersurface, here the mass-shell constraint.

Next we calculate the hamiltonian, which is trivially vanishing:

$$H = p_\mu \dot{q}^\mu - L = 0.$$

This means in this system, evolution is simply a gauge transformation and there are no dynamics.

We consider the phase space to be  $\mathcal{M} = T^*\mathbb{R}^{D+1}$  with poisson brackets among  $q^a$  and  $p_a$ . Our constraint is the mass-shell constraint, and we choose  $\mathcal{H}_{kin} = L_2(\mathbb{R}^{D+1}, d^{D+1}x)$ ,  $\mathcal{D}_{kin} = \mathcal{S}(\mathbb{R}^{D+1})$  and  $\mathcal{D}_{kin}^* = \mathcal{S}^*(\mathbb{R}^{D+1})$ . This system is rather similar to the momentum one, so we work in the momentum representation again to get

$$l(\hat{C}\Psi) = 0 \implies l((-\partial^2 + m^2)\Psi) = 0 \implies l((-p_0^2 + p^2 + m^2)\Psi) = 0.$$

We assume the solution is of the form  $l_f = \delta(-p_0^2 + p^2 + m^2)f(p_0, p_i)$ . Applying the same physical inner product as in 1, we can get

$$\langle l_{f_1}, l_{f_2} \rangle = \int l_{f_1}^* f_2(p_i) d^D p.$$

that is to say the physical hilbert space is  $\mathcal{H}_{phys} := L_2(\mathbb{R}^2, d^D p)$ . The dirac observables are  $Q^a = q^a - q^0 \frac{p_a}{\sqrt{m^2 + p^2}}$ . It is easy to verify that for every  $q^0$ ,  $[Q^a, C] = 0$ . This is another sign that coordinate time in relativistic systems are gauge, and for simplicity reasons we set  $q^0 = 0$ .

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## 2.4 Maxwell Theory

As we all know, the lagrangian for a free maxwell theory is  $\mathcal{L} = -1/4 F_{\mu\nu} F^{\mu\nu}$ . We define  $E^a = \dot{A}_a - \partial_a A_0$ , and  $B^a = \epsilon^{abc} \partial_b A_c$ . The lagrangian is therefore

$$L = \int d^3x \frac{1}{2} (E^a E_a - B^a B_a)$$

the generalized coordinates in this case are  $A^\mu$ . If we compute the conjugate momenta of  $A^\mu$ , we find

$$\begin{cases} \frac{\partial L}{\partial A_0} = 0 =: \pi^0, \\ \frac{\partial L}{\partial \dot{A}_a} = E^a =: \pi^a. \end{cases}$$

and the hamiltonian to be

$$H = \int d^3x \left( \pi^0 \dot{A}_0 + \pi^a \dot{A}_a - \mathcal{L} \right) = \int d^3x \frac{1}{2} (\pi^2 + B^2) + \pi^a \partial_a A_0.$$

For the term  $\pi^a \partial_a A_0$ , we perform integration by parts:  $\pi^a \partial_a A_0 = -A_0 \partial_a \pi^a$ . The time evolution of  $\pi^0$  should be 0, which leads to the poisson bracket  $\{\pi^0, H\} = -\delta H / \delta A_0 = \partial_a \pi^a = \partial_a E^a = 0$ . But this is exactly the Gauss constraint!  $A_0$  has lost all it's dynamics under this constraint, but only a lagrange multiplier.

Now we demonstrate that the Gauss constraint does indeed generate the  $U(1)$  gauge transformation. We consider the integration of  $\partial_a E^a$  multiplied by some function  $\lambda(x)$ :

$$C(\lambda) = \int d^3x \lambda(x) \partial_a E^a = - \int d^3x (\partial_a \lambda) E^a$$

Varying  $A$  gives

$$\delta A(x) = \{A_a(x), C(\lambda)\} = -\partial_a \lambda(x),$$

which is the standard  $U(1)$  gauge transformation. Similarly, we can see that  $E$  is gauge invariant.

In order to do quantization, we first apply the Helmholtz decomposition on  $A_a$  and  $E^a$ :

$$\begin{cases} A_a = A_a^L + A_a^T, \\ E^a = (E^L)^a + (E^T)^a. \end{cases}$$

where  $\nabla \cdot A^T = 0$ ,  $A_a^L = \partial_a \phi$  and  $\nabla \cdot E^T = 0$ ,  $(E^L)^a = \partial_a \psi$ . Now the Gauss constraint becomes  $C = \partial_a E^a = \partial_a (E^L)^a = \nabla^2 \psi = 0$ . Therefore, the gauge condition suggests that the dirac observables are the transversal parts of  $A$  and  $E$ .

Now we do the quantization. Consider the three corresponding massless scalar fields of  $A_a$ .  $\mathcal{H}_{kin}$  is the Fock space generated by  $\hat{z}_a^\dagger(k)$  and  $\hat{z}_a(k)$ , the creation and annihilation operators respectively. Under fourier transform, the constraint can be written as  $\hat{C} = k_a E^a(k)$ , which leads to  $k^a \hat{z}_a(k) |\Psi\rangle_{phys} = 0$ . Next we introduce polarized vectors  $e_a(k)$ . But because by the gauss constraint  $E^a$  only consist of components parallel to  $k$ , the constraint equation is simply  $\hat{z}_3(k) |\Psi\rangle_{phys} = 0$ . This means physical states are states without longitudinal excitations, therefore  $\mathcal{H}_{phys}$  is the Fock space spanned by  $\hat{z}_1^\dagger(k)$  and  $\hat{z}_2^\dagger(k)$ . This is why there are only two tranverse polarization states of a photon.

In the next part, we will use the ADM formulation to construct the phase space of General Relativity.